General Biology 1

First Semester

Supplementary Materials

Module 7

Meiosis: The Importance of Crossing Over

Raymart F. Redita
Author

What I Know

Directions: Read each situation carefully and choose the best answer.

1. During which stage of meiosis does crossing over most likely occur?

a. Prophase I
b. Metaphase II
c. Anaphase I
d. Telophase II

2. What is the main purpose of crossing over?

a. To reduce chromosome number
b. To produce identical cells
c. To exchange genetic material between homologous chromosomes
d. To separate sister chromatids

3. Which structures are involved in crossing over?

a. Sister chromatids of the same chromosome
b. Homologous chromosomes
c. Ribosomes
d. Cell membrane

4. What is the result of genetic recombination?

a. Cells with identical genetic information
b. Increased genetic variation among offspring
c. Faster cell division
d. Reduction of cytoplasm

5. If crossing over did not occur during meiosis, what would most likely happen?

a. Chromosomes would not replicate
b. Offspring would have less genetic variation
c. Cells would not divide
d. DNA would be destroyed

What I Need to Know

This supplementary module is designed to help you understand the phases of meiosis and the importance of crossing over. In this lesson, you will explore the different phases of meiosis and how recombination works in increasing genetic diversity among offspring

Lesson 7 Meiosis

The Importance of Crossing Over

Meiosis is a specialized type of cell division that produces gametes (sex cells such as sperm and egg) with half the number of chromosomes of the original cell. Meiosis occurs in two consecutive divisions: Meiosis I and Meiosis II. Homologous chromosomes pair up and exchange genetic material through crossing over during prophase I. The outcome of meiosis is four genetically unique haploid cells, each different from the parent cell and from one another. This genetic variation is mainly due to crossing over and the independent assortment of chromosomes. Meiosis plays a crucial role in sexual reproduction, contributing to the diversity seen among organisms.

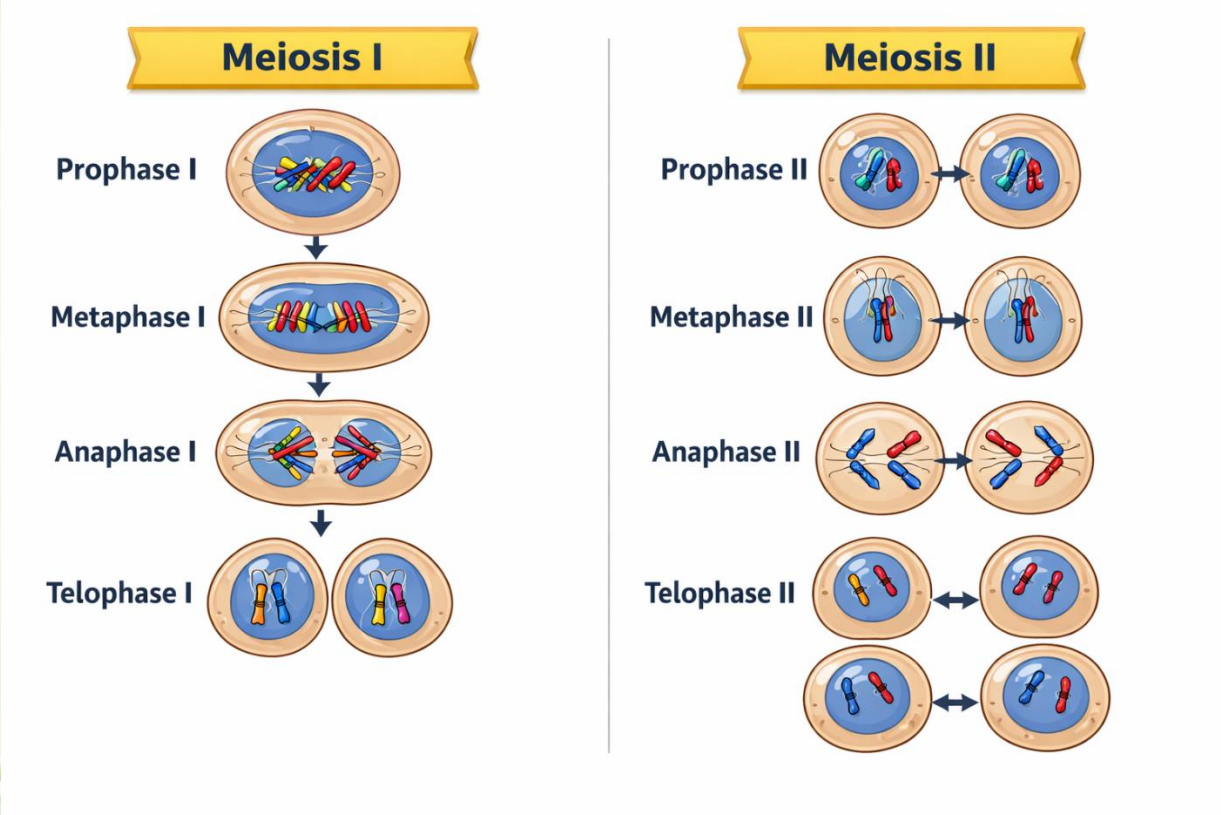
After going through this supplementary module, you are expected to:

1. describe the process of crossing over during prophase I of meiosis, including how homologous chromosomes exchange genetic material;
2. explain how recombination produces new combinations of alleles and contributes to genetic variation in gametes; and
3. analyze the significance of crossing over and recombination in increasing genetic diversity among offspring.

What's In

Activity 1: "Meiosis in Action: Stage by Stage Challenge"

Directions: Using the diagram of meiosis I and II, identify which phase a specific event occurred in the cell division.



1. Exchange of genetic material between homologous chromosomes increases variation.
   Answer: _____
2. Homologous chromosome pairs line up in the middle of the cell.
   Answer: _____
3. Each chromosome moves to opposite poles (sister chromatids remain attached).
   Answer: _____
4. Chromatids are pulled apart to opposite poles, becoming individual chromosomes.
   Answer: _____

5. Final division results in four genetically unique gametes.
   Answer: _____
6. Allows spindle fibers to interact with chromosomes.
   Answer: _____
7. Cell divides, reducing chromosome number by half.
   Answer: _____
8. Chromosomes are arranged randomly, contributing to genetic diversity.
   Answer: _____
9. Similar to mitosis, chromosomes line up individually.
   Answer: _____
10. No DNA replication occurs before this stage.
    Answer: _____

Meiosis I

1. Prophase I

- Both the nuclear envelope and the nucleolus begin to disappear.
- The centriole forms spindle fibers.
- Homologous chromosomes undergo pairing or the process of synapsis. The homologous chromosomes have the same length and position of the centromere. These homologous chromosomes came from the father and the mother. One paired homologous chromosome is composed on four sister chromatids, which is called a tetrad.
- Crossing over between the two non-sister chromatids takes place. Crossing over is the swapping of genetic material between non-sister chromatids along the point called chiasmata. As a result, genetic recombination takes place and said to be accountable for genetic variation.

2. Metaphase I

- The paired homologous chromosomes have moved toward the metaphase plate or at the center of the cell.
- Spindle fibers are fully formed and have increased in number.
- Kinetochores, protein complexes outside the centromeres are present and attached to the spindle fibers.

3. Anaphase I

- Homologous chromosomes separate and begin moving to the opposite poles.
- Since there are 46 chromosomes in human, 23 will move to the different poles. This means that only half of the number of chromosomes will remain in each cell.

4. Telophase I

- Finally, the two daughter cells completely divide.
- There is an equal number of chromosomes found in each daughter cells.
- The nuclear envelope and the nucleus reappear, and the chromosomes become less visible in both cells.

Interkinesis

Interkinesis is a process similar to interphase between mitotic divisions except the replication of DNA since chromosomes are already duplicated. This is considered as a short pause between the two phases of meiosis.

Meiosis II

5. Prophase II

- Both the nuclear envelope and the nucleolus dissolve or disappear.
- The chromatids begin thickening and shortening.
- The centrioles move to the opposite poles and the spindle fibers arrange to prepare for the next phase.

6. Metaphase II

- The two kinetochores of each centromere bind to the spindle fibers from the opposite poles.
- The chromosomes move until they are on the metaphase plate.

7. Anaphase II

- The centromeres separate, allowing the microtubules attached to the kinetochores to pull and move to the poles.
- The sister chromatids become sister chromosomes after moving towards the end of the opposite poles.

8. Telophase II

- The spindle fibers disappear.
- The nuclear envelope starts to form around the daughter chromosomes.
- At Telophase II, four haploid daughter cells are the end result where these daughter cells are genetically distinct from the parent cell.

What's New

Activity 2: "Crossing Over: The Genetic Exchange Lab"

Directions: Read carefully the scenario given and accomplish the task intended to complete this activity.

Activity Scenario

You are genetic researchers investigating why siblings look different even if they have the same parents. Your task is to model what happens during Prophase I of meiosis, specifically how chromosomes exchange segments and create variation.

Instructions:

1. Use two different colored strips (one from each parent).
2. Draw 3–5 labeled genes (A, B, C, etc.) on each strip.
3. Pair the homologous chromosomes side by side.
4. Cut matching segments from each chromosome.
5. Swap the segments and tape them to the opposite chromosome.
6. Compare the new chromosomes with the original ones.
7. Identify the new gene combinations formed.

Paste here your chromosome model before crossing over

Paste here your chromosome model after crossing over

Guide Questions:

1. What is crossing over and when does it occur?

   _____

   _____

2. How does crossing over increase genetic variation?

   _____

   _____

3. What might happen if crossing over did not occur?

   _____

   _____

4. Why is genetic variation important for organisms?

   _____

   _____

What's More

Activity 3: "The Power of Crossing Over"

Directions: Read carefully the scenario given and accomplish the task intended to complete this activity.

Activity Scenario

A population of insect's lives in a forest where conditions suddenly change due to climate shifts. A new disease spreads, and many insects begin to die. Scientists observe that only some individuals survive, and these survivors have slight genetic differences that make them resistant to the disease.

Upon investigation, scientists discover that these differences are linked to genetic variation produced during meiosis, particularly through crossing over and recombination. They now want to understand how this process helped certain individuals survive while others did not.

Guide Questions:

1. What is crossing over, and during which stage of meiosis does it occur?

   _____

   _____

   _____

2. How does crossing over lead to new combinations of genes in gametes?

   _____

   _____

   _____

   _____

3. Why might individuals with more genetic variation have a higher chance of survival?

   _____

   _____

   _____

4. How did crossing over contribute to the survival of some insects in the scenario?

_____

_____

_____

5. What could happen to a population if crossing over did not occur during meiosis?

_____

_____

_____

To experience **MORE** interactivity in this module, scan this QR code.

What I've Learned

Directions: Encircle the letter of the correct answer.

1. A population of plants shows very little variation and becomes vulnerable to a sudden disease outbreak. Which explanation best describes how crossing over could have helped prevent this situation?

a. It increases the number of chromosomes in each cell
b. It creates identical copies of parental genes
c. It produces new combinations of alleles that may include disease resistance
d. It speeds up the process of cell division

2. During meiosis, two homologous chromosomes exchange segments unevenly. What is the most likely effect of this event on the resulting gametes?

a. All gametes will be genetically identical
b. Gametes will contain new allele combinations
c. Chromosome number will double
d. No genetic change will occur

3. If crossing over did not occur during meiosis, which outcome would most likely be observed in offspring?

a. Increased mutation rates
b. Reduced genetic diversity among offspring
c. Faster reproduction
d. More dominant traits expressed

4. Two siblings inherit genes from the same parents but show different traits. Which process best explains this difference?

a. DNA replication
b. Mitosis
c. Crossing over during Prophase I
d. Cytokinesis

5. A scientist compares two groups of organisms: one with normal meiosis and one where crossing over is inhibited. Which conclusion is most valid?

A. Both groups will have equal genetic variation
B. The group without crossing over will show greater diversity
C. The group with normal meiosis will have more varied offspring
D. Crossing over has no effect on inheritance